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Research Article

Enhancing Phishing Detection: A Machine Learning Approach to Predicting Malicious Emails, URLs, and SMS Messages

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The rapid digitization of financial service companies and organizations has heightened concerns over data security, particularly regarding unauthorized access to personal, financial, and password information. Among the various cyber threats, phishing attacks remain a pervasive and evolving challenge, exploiting deceptive emails, URLs, and SMS messages to illicitly harvest sensitive user data. In response, this research investigates a machine learning and deep learning–based framework for comprehensive phishing detection across multiple phishing datasets. We integrated diverse datasets, including phishing emails from Kaggle, malicious SMS messages from Mendeley, and both phishing and legitimate URLs from Kaggle and OpenPhish, to train the proposed models. Leveraging this carefully preprocessed and heterogeneous dataset, we systematically evaluate and compare the performance of nine state-of-the-art machine learning and deep learning models. Furthermore, we propose an ensemble approach that synergizes the strengths of these models to enhance detection accuracy. Our experimental results demonstrate exceptional performance, with support vector machine (SVM) and bidirectional gated recurrent unit (BiGRU) models achieving the highest detection accuracies of 99.91% and 99.75%, respectively. The findings indicate that deep learning architectures generally outperform traditional machine learning techniques in identifying phishing attempts. This work not only provides a robust, multimodal phishing detection solution but also offers valuable insights for strengthening cybersecurity defenses in an increasingly digital world.

Keywords: cybercrime; cybercriminals; cybersecurity; deep learning; machine learning; phishing attacks; phishing detection

1. Introduction

The internet has become an essential part of modern life, offering vast opportunities for individuals and businesses worldwide. However, its growing influence and accessibility have also introduced new online threats, with cybercriminals becoming increasingly sophisticated. Among these threats, phishing stands out as a particularly common and dangerous form of cyberattack. Phishing involves deceptive attempts to steal personal information by exploiting human psychology and social engineering techniques. Attackers often impersonate trusted entities, such as financial institutions, government agencies, or popular service providers, making these scams highly effective and difficult to detect. These

attacks jeopardize victims' financial stability and reputational integrity, underscoring the urgent need for robust cybersecurity measures, heightened awareness, and proactive defense strategies to mitigate the risks posed by such malicious activities [1]. An overview of a typical phishing attack is illustrated in Figure 1.

Figure 2 shows the example of a phishing email. You received an email pretending to come from Apple, when in fact, the sender's address was "supportapple.com" instead of the real email address, "support@apple." The slight variation in the domain name is an obvious red flag. Likewise, you might receive an email from "secure.bankupdate@banksecure.com," subject line "URGENT: Security Alert for Your Account." It begins with a generic opening line: "Dear Customer" and advises, "Your account has been compromised. Your account will be suspended unless you reset your password urgently." It has a link that reads "Reset Password"; when the cursor hovers over it, it takes you to "http://securebank-loginreset.com," and not the bank's domain. Both examples share common signs of phishing, including dubious sender IDs, generic salutations, urgent tones, unusual requests, and falsified links, and serve as textbook phishing scams [2].

Phishing attacks continue to pose significant risks to organizations and individuals worldwide, resulting in substantial financial losses and damage to reputations. These attacks exploit human vulnerabilities by using deceptive techniques to gain unauthorized access to sensitive information, systems, and infrastructure. Several notable incidents highlight the severe consequences of phishing. For example, in 2013, a Lithuanian hacker impersonated a legitimate vendor and successfully defrauded Google and Facebook of over $100 million through fake invoices. Phishing emails compromised credentials at Bangladesh Bank in 2016, resulting in a loss of $81 million via the SWIFT financial system. Critical infrastructure has also been targeted; a spear-phishing attack on Ukraine's power grid in 2015 caused large-scale power outages, marking one of the first successful cyberattacks against essential services. Supply chain vulnerabilities present another major concern, exemplified by the 2013 Target breach, which originated from a phishing attack on a third-party vendor and exposed personal data for approximately 110 million customers. Educational institutions are not immune either; in 2018, the Australian National University suffered a sophisticated phishing attack that compromised decades of staff and student personal data. Financial institutions remain prime targets, as demonstrated by India's Cosmos Bank losing $13.5 million after cybercriminals exploited phishing to infiltrate ATM systems. Furthermore, the COVID-19 pandemic intensified phishing risks, including targeted attacks on the World Health Organization, where attackers pretended to be officials to steal credentials and disrupt pandemic response efforts. These examples underscore the persistent and evolving threat of phishing, illustrating the dire need for robust cybersecurity measures. Organizations and individuals must adopt proactive strategies such as establishing stringent security policies, providing regular awareness training, and implementing multifactor authentication to effectively safeguard against these malicious attacks [3–6].

Phishing detection has significantly progressed with the adoption of various machine learning (ML) algorithms such as random forest, Naive Bayes, K-nearest neighbors (KNNs), and support vector machines (SVMs). These traditional methods are particularly effective at analyzing structured features such as URL characteristics, email metadata, and HTTP headers. Recently, deep learning (DL) models, including convolutional neural networks (CNNs) and long short–term memory (LSTM) networks, have further advanced detection by automatically extracting complex, high-dimensional features from unstructured data sources such as HTML email bodies, webpage content, and URL sequences. Combining these approaches into ensemble or hybrid frameworks has boosted both detection accuracy and robustness. Despite these advancements, phishing remains a dynamic target due to constantly evolving techniques such as dynamically generated URLs, domain spoofing, and artificial intelligence–powered attacks that can evade even sophisticated models. Furthermore, phishing datasets often contain many irrelevant or redundant features, which can degrade model performance and increase computational demands. To overcome these challenges, it is essential to apply effective feature selection and engineering strategies that streamline input data, improve model efficiency, and reduce processing time [7, 8].

As phishing detection systems become increasingly complex, there is a growing need for solutions that effectively balance speed, accuracy, and scalability, particularly for real-time applications. To meet these challenges, this research proposes an integrated architecture that combines ML, DL, and feature optimization techniques. By leveraging advanced feature extraction methods alongside socio-technical learning models and incorporating real-time deployment strategies, this approach aims to develop a robust and scalable framework. Guided by a research-driven methodology, the goal is to build a comprehensive system capable of adapting to and countering the constantly evolving landscape of phishing threats.

In this research, we selected a diverse set of ML and DL models to comprehensively evaluate phishing detection performance across multiple integrated diverse datasets [9–11]. Traditional classifiers such as SVM, random forest, and KNNs were chosen for their proven effectiveness with handcrafted features and their ability to capture different data distributions and nonlinear relationships. DL architectures, including LSTM, bidirectional gated recurrent unit (GRU) (BiGRU), and CNN-LSTM hybrids, were employed to leverage their strengths in modeling sequential and contextual patterns inherent in phishing emails, URLs, and SMS messages. In addition, the transformer-based bidirectional encoder representations from transformers (BERT) model was included for its superior capability in understanding semantic nuances and subtle linguistic cues, which are critical for detecting sophisticated phishing attempts. This combination of models enables a thorough comparison of traditional and advanced approaches, highlighting their respective advantages in phishing detection.

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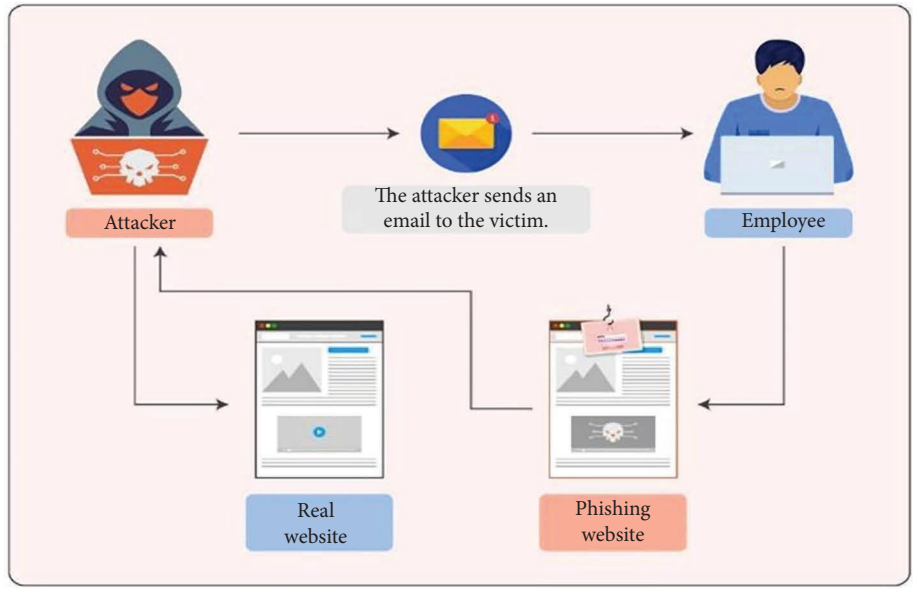


Figure 1: Phishing attack overview.

This research makes several significant contributions to the field of phishing detection and cybersecurity:

1. We address the reality of multichannel phishing by integrating multiple, publicly available datasets, including phishing emails from Kaggle, malicious SMS messages from Mendeley, and comprehensive collections of phishing and legitimate URLs from Kaggle and OpenPhish. This results in a heterogeneous, real-world training foundation that reflects the variety of phishing techniques encountered today.
2. To ensure thoroughness, we systematically evaluate and compare nine state-of-the-art ML and DL models. These include both standard classifiers and more complex models, giving us a strong way to measure and understand how well different methods can detect phishing in various types of data.
3. Beyond individual model performance, we propose a novel ensemble approach that combines the predictive strengths of selected models. This ensemble achieves improved detection accuracy, demonstrating the practical value of integrating complementary algorithms for real-world applications.

The rest of this paper is organized as follows. We discuss closely related work in Section 2. Section 3 introduces the background of our work. Section 4 introduces the proposed methodology and gives an overview of our approach. Section 5 presents the performance analysis of our proposed model. We conclude the paper with future work in Section 6.

2. Related Work

Phishing is a widespread and damaging form of cybercrime in which attackers impersonate trusted entities to steal sensitive personal information, such as passwords and credit card details, resulting in data breaches and significant financial losses. Unlike the dramatic scams often depicted in popular media, phishing operates as a stealthy and pervasive threat that takes many forms, including traditional email phishing, spear-phishing (targeted, personalized attacks), smishing (fraudulent text messages), vishing (voice call scams), and more sophisticated tactics such as whale phishing (targeting high-level executives) and angler phishing (posing as customer service on social media platforms). Cybercriminals exploit digital vulnerabilities, including weak identification protocols in emerging services such as Buy Now, Pay Later (BNPL), and increasingly prey on vulnerable populations such as seniors, who face the risk of digital elder abuse [12].

The threat landscape has grown more complex with the rise of ransomware, which caused estimated losses of $590 million in the first half of 2021 alone, and romance scams that resulted in losses of $133 million during the same period. Cryptocurrency fraud has similarly surged, with approximately 7000 victims losing $80 million over six months. Alarmingly, only 16% of organizations report having avoided recent cyberattacks, while 70% anticipate disruptions from email-based threats. Despite ongoing cybersecurity initiatives, employee confidence in detecting phishing attempts remains low: just 45% feel capable of identifying phishing emails, and even fewer, 34%, can recognize smishing or vishing scams. Industries such as healthcare, manufacturing, finance, and education are particularly exposed, facing frequent ransomware, data theft, and targeted attacks. The global financial impact of cybercrime is staggering, with damages reaching an estimated $6 trillion in 2021 and projected to rise to $10.5 trillion annually by 2025. The advertising industry loses approximately $51 million daily to fraud, while ransomware attacks are expected to occur every two seconds by 2031, costing an estimated $265 billion each year. As cybercriminals continue to refine their tactics, leveraging artificial intelligence and other emerging technologies, businesses and individuals

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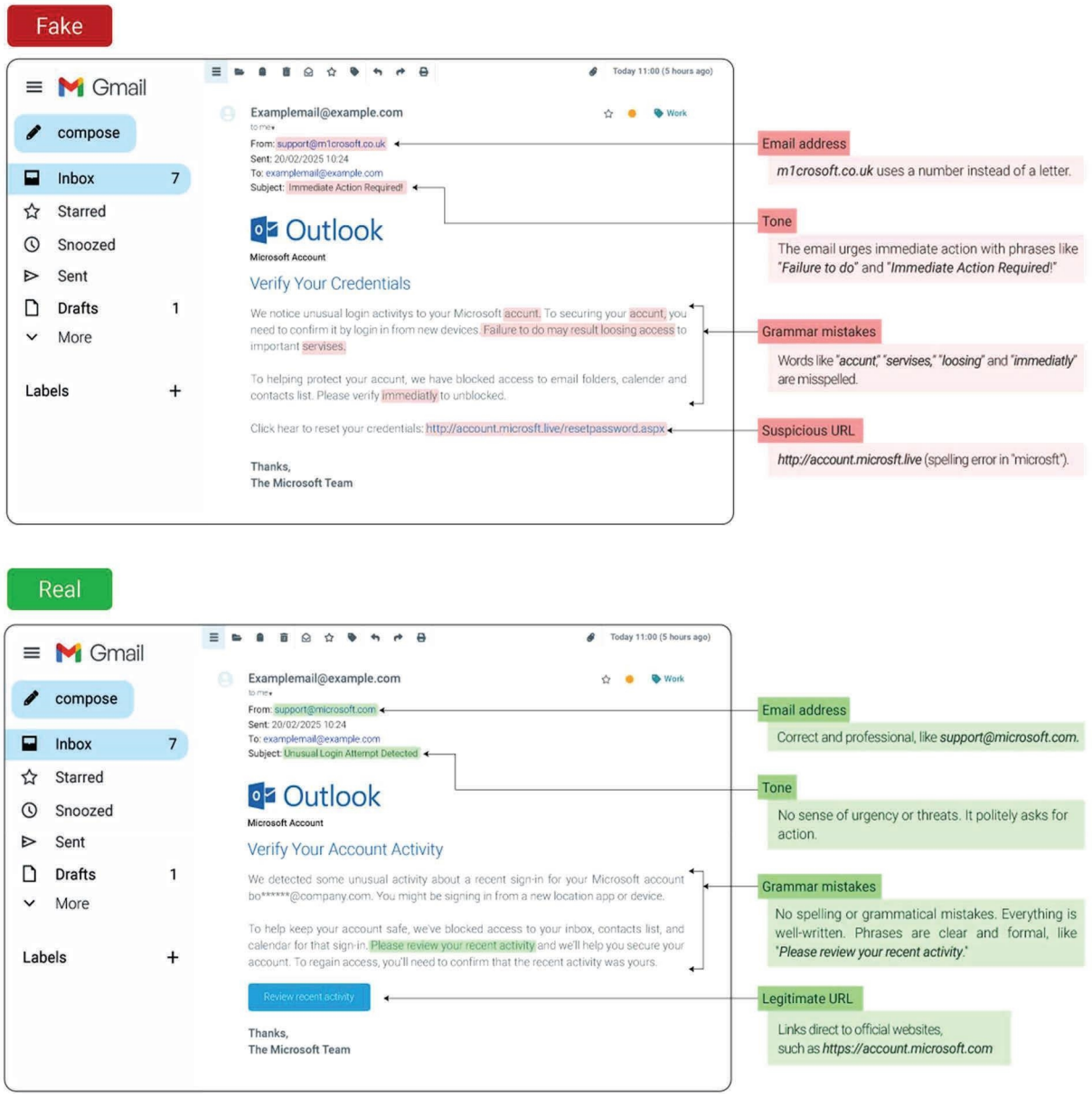


Figure 2: Example of a phishing email.

must prioritize cybersecurity awareness, deploy advanced detection tools, and implement robust authentication measures to effectively mitigate these evolving threats [13].

During the third quarter of 2024, social media platforms emerged as the predominant target for phishing attacks, accounting for 30.5% of incidents globally, as illustrated in Figure 3. Following closely, web-based software services and webmail platforms together constituted approximately 21.2% of reported phishing attacks. Financial institutions also faced considerable threats, representing 13% of the total phishing incidents during this period. This distribution underscores the diverse range of sectors vulnerable to cyber threats and points out the importance of enhanced security measures across all digital platforms [12].

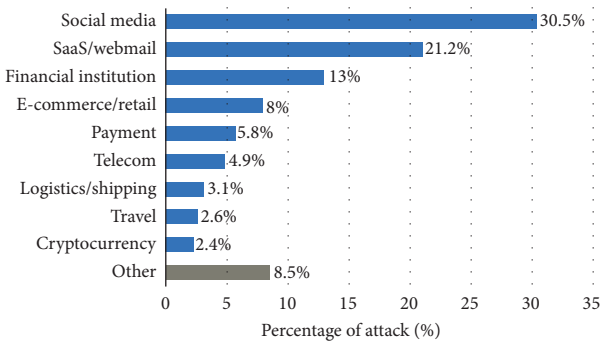


Figure 3: Phishing most targeted industry sectors worldwide Q3 2024 [12].

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The British Airways (BA) breach, often cited as a cautionary tale, demonstrates the real-world risks and consequences of a URL redirect attack, particularly when security measures are reactive rather than preventative. The attacker edited a JavaScript file on the BA booking website, inserting code to redirect payment information to a malicious lookalike domain, baways.com, controlled by the attacker. The redirect was active for approximately 15 days, diverting sensitive customer data (names, addresses, payment card numbers, and security codes) as users entered them on the legitimate site. Nearly 430,000 customers and staff had personal and payment information compromised, many exposed to further fraud or scams. BA was fined £20 million (down from an initial £183 million) for failing to protect customer data adequately under the GDPR. The regulator's comments made it clear that the lack of organizational and technical controls, not the hack alone, was the main basis for the penalty. The breach resulted in loss of consumer trust and regulatory scrutiny and highlighted the need for integrating cybersecurity deeply into every business function. The BA incident underscores how URL redirect vulnerabilities, often dismissed as low-level threats, can be leveraged for large-scale data theft when broader business security culture and controls are insufficient. Effective, organization-wide prevention strategies, ongoing technical remediation, and regular staff training are essential to avoid similar attacks [13].

In recent years, the landscape of phishing detection has undergone a dramatic transformation. The field has largely moved away from traditional detection techniques, such as blacklist-based and rule-based systems, toward state-of-the-art solutions powered by ML and DL algorithms. This significant shift has been primarily driven by the inherent drawbacks of static strategies, specifically their inability to adapt to novel phishing methodologies and the escalating demand for real-time detection capabilities. Table 1 provides a comprehensive overview of existing ML- and DL-based phishing attack detection techniques, highlighting their respective methodologies, datasets utilized, evaluation metrics, and reported results.

3. Strategic Model Selection for a Generalized Phishing Detection Framework

Our research's primary objective was to develop a generalized and practical phishing detection framework that could effectively identify threats across three distinct communication channels: emails, URLs, and SMS messages. These channels have special difficulties due to their differing structures, data modalities, and semantic patterns. To address this, we strategically selected nine ML and DL models, ensuring each was well-suited to a specific data type and learning paradigm.

3.1. Models for Structured URL Data. Phishing URLs are rich in structured, nonlinear features, such as domain age, character length, and the presence of special symbols. For these types of data, tree-based models are particularly effective.

XGBoost was chosen for its ability to capture complex feature interactions, which is crucial for identifying subtle and sophisticated URL patterns.

Random forest was included for its robustness against noise and its ability to provide interpretable feature importance scores, offering insights into which URL characteristics are most indicative of a threat.

3.2. Models for Unstructured Textual Data. Phishing emails and SMS messages consist of unstructured text, where malicious intent is embedded in linguistic tone and phrasing. We used models designed to handle high-dimensional textual data.

SVM excels at creating robust decision boundaries in high-dimensional, sparse feature spaces, making it ideal for features derived from text, such as term frequency–inverse document frequency (TF-IDF) vectors.

KNNs served as a simple, nonparametric baseline, helping us evaluate how well phishing content could be classified based on simple feature similarity alone.

3.3. Models for Sequential and Temporal Patterns. Phishing content often contains temporal dependencies and sequential cues that can reveal malicious intent. We incorporated a suite of recurrent neural networks (RNNs) to model these properties.

LSTM was used to capture long-range dependencies, which are critical for detecting subtle cues in lengthy phishing emails.

GRU, a more computationally efficient variant, was included to assess whether similar performance could be achieved at a reduced computational cost, a key consideration for real-time applications.

BiGRU was selected for its ability to process text in both directions, capturing contextual information from both the start and end of a message. This proved particularly valuable for detecting phishing indicators that can appear anywhere in an SMS message.

3.4. Advanced Hybrid and Contextual Models. To push beyond traditional RNNs and leverage the latest advancements in natural language processing (NLP), we included two advanced models.

CNN-LSTM combines CNNs for localized pattern recognition (e.g., specific phishing phrases), with LSTMs for global sequential understanding. This hybrid approach is highly effective for SMS messages, which often have both localized and temporal patterns.

BERT, a state-of-the-art model, was used to evaluate whether a pretrained, context-aware model could outperform models trained from scratch, especially when generalizing across different text domains.

Our results justify this diverse and strategic selection. While all models performed well, our nuanced evaluation of the RNN variants confirmed their distinct strengths. For example, the BiGRU model, despite a small difference in overall accuracy, consistently achieved the highest precision

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Table 1: Summary of related works on phishing detection.

Ref.	Model	Dataset	Performance evaluation	Training and testing accuracy
[13]	NLP and DL (LSTM, BiLSTM, GRU, BiGRU)	Phishload database with 10,373 records of phishing and ham pages.	Accuracy, precision, recall, *F*1-score	Best: BiGRU with 97.39% accuracy. Mean accuracy across models: 96.7%
[14]	Enhanced K-means clustering	Public datasets of 2000, 7000, and 10,000 samples from a repository	Accuracy (89.2% on 2000 samples), precision, recall, *F*1-score	Improved accuracy compared to kernel K-means: 89.2% (enhanced) vs. 51.5% (traditional)
[15]	CNN	PhishTank dataset with 10,000 phishing URLs and 10,000 legitimate URLs	Accuracy, false positive rate (FPR), area under the curve (AUC), precision, recall	CNN model achieved 98.77% accuracy. Compared: KNN (87%), NLP-based (97.98%), RNN (97.4%), RF (94.26%)
[16]	CNN, LSTM, LSTM-CNN	PhishTank, Common Crawl, Alexa datasets	Accuracy, recall, *F*1-Score, area under the curve (AUC)	CNN: 99.2%, LSTM-CNN: 97.6%, LSTM: 96.8%
[17]	Neural network (NN), Naïve Bayes, AdaBoost	Kaggle dataset (11,000+ websites, 30 features)	Confusion matrices, precision, recall, and accuracy	NN: 90.23%, Naïve Bayes: 92.97%, AdaBoost: 95.43%
[18]	CatBoost, AdaBoost, random forest, SVM, NLP	Publicly available labeled URL dataset	Compared models based on accuracy, runtime latency	Random forest: 98.3%, SVM: high, overall accuracy
[19]	Artificial neural network	Custom phishing dataset with features extracted from real-world examples	ROC, precision–recall curves; feature selection and ANN optimization	Achieved testing accuracy above 95% depending on selected features
[20]	ML, DL, hybrid learning, scenario-based techniques	PhishTank, Alexa, UCI Machine Learning Repository datasets with ~50,000 instances	*F*1-score, precision, and recall for detection across ML and hybrid approaches	Accuracy varied from 89.75% to 99.2% depending on the approach and the dataset used
[21]	DT, NB, SVM, MLP	The dataset includes 500 phishing and 500 legitimate websites, collected from PhishTank (phishing) and Alexa (legitimate). Features include 25 URL, HTML, and HTTP attributes	Evaluated using accuracy, precision, recall, *F*1-score, and classification time. DT with URL + HTTP features achieved the best *F*1-score (99%) and accuracy (99%)	DT: train 99%, test 99%; NB: train 67%, test 67%; SVM: train 98%, test 97%; MLP: train 98%, test 96%
[22]	AI-generated phishing email detection	The dataset consists of 865 AI-generated phishing emails (DeepAI), a dataset of Enron emails of size 500, 000, a dataset of Ling-Spam emails of size 2412, and a corpus of Nigerian phishing emails. Emails generated by AI, studied by length, body structure, and style	Evaluated accuracy, precision, recall, *F*1-score, and classification time. DT achieved the best *F*1-score (99%) on URL + HTTP features. NB showed the lowest accuracy (67%)	Ensemble: train 99.5%, test 99.5%; Naive Bayes: train 99.3%, test 99.3%; UDAT: train 98%, test 98%; LSTM DNN: train 99%, test 99%
[23]	Fusion, RF, SVM, KNN, logistic regression	The dataset contains 15,000 samples, of which half are phishing and the other half are legitimate, that are obtained from the PhishTank and OpenPhish and use URL and HTML-based characteristics. Some of the features extracted are URL length, special characters, HTML tags, and HTTPS	Evaluated accuracy, precision, recall, *F*1-score, confusion matrix. Best accuracy (99.5%) achieved with the ensemble approach. Style-based descriptors identified AI-generated phishing emails effectively	Fusion: train 98.7%, test 98.5%; RF: train 98%, test 97%; SVM: train 97.5%, test 96%; KNN: train 96%, test 95%; logistic regression: train 93%, test 92%
[24]	RF, XGBoost, DT, NB	The dataset contains 51,316 legitimate websites and 36,173 phishing websites sourced from PhishTank. Features include 73 URL-based attributes such as URL length, domain age, and HTTPS presence	Evaluated accuracy, recall, precision, *F*1-score, and runtime efficiency. GA-based feature selection reduced features to 44 while improving recall and accuracy. RF achieved the best recall (89.05%)	RF: train 92.93%, test 92.8%; XGBoost: train 91.53%, test 91.5%; DT: train 88.83%, test 88.7%; NB: train 85.4%, test 85.3%

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Table 1: Continued.

Ref.	Model	Dataset	Performance evaluation	Training and testing accuracy
[25]	1D Convolutional neural network (1D CNN)	Large-scale dataset including 200,000 phishing URLs and 200,000 legitimate URLs from PhishTank, UNB, and Alexa repositories. Features include URL structure and domain-related attributes	Evaluated using accuracy, precision, recall, and *F*1-score. Achieved superior accuracy (99.7%) with balanced precision and recall metrics	Training accuracy: 99.7%; Testing accuracy: 99.3%
[26]	Multiple machine learning techniques	1. Phishing email collection: 11,727 phishing and legitimate samples with 22 features. 2. Phishing legitimate full: 10,000 samples with 50 features. 3. Spam or not spam dataset: 3000 samples with text-only features	Accuracy, precision, recall, and *F*1-score. Boosted decision tree achieved the highest accuracy (100%) on the second dataset. Neural network and averaged perceptron performed best on the third dataset (97.7%)	Dataset 1: train 99%, test 88.8%; Dataset 2: train 100%, test 100%; Dataset 3: train 97.7%, test 97.7%
[27]	Deep learning–based phishing detection	Data sources: PhishTank, OpenPhish, Alexa top sites, WHOIS data, third-party website information, email content, and URL attributes (e.g., domain info, query paths)	Evaluated accuracy, precision, recall, *F*1-score, and AUC. Hybrid models achieved superior accuracy (98%–99%) across datasets	Average training accuracy: 98%–99%, testing accuracy: 98.5%
[28]	Supervised and deep learning–based framework. SVM, LSTM, NB	Collected from the CSDMC_SPAM Dataset, with over 4000 phishing and legitimate emails. Features extracted include email header, body, and metadata such as sender/receiver attributes and timestamp	Evaluated using precision, recall, *F*1-score, accuracy, and confusion matrix. SVM achieved the highest accuracy (99.62%), followed by LSTM (98%) and NB (97%)	SVM: train 99.7%, test 99.62%; LSTM: train 98%, test 98%; NB: train 97%, test 97
[29]	1D CNN architecture.	The dataset is composed of 200k phishing URLs and 200k legitimate URLs, sourced from PhishTank, UNB, and Alexa. Features included URL structure, domain information, and embedded tokens	Evaluated accuracy, recall, precision, and *F*1-score. Achieved the highest accuracy (99.7%) among compared methods.	Training accuracy: 99.7%; testing accuracy: 99.3%
[25]	Phishing website detection framework. DT, RF, XGBoost, AdaBoost	The dataset includes 95,911 phishing and legitimate URLs, sourced from Kaggle and proprietary collections. Features include URL structure, special characters, HTTPS presence, and domain-based attributes	Evaluated using accuracy, precision, recall, and *F*1-score. Decision tree achieved the highest accuracy (95.5%), with random forest and XGBoost closely following	Decision tree: train 100%, test 95.5%; random forest: train 95%, test 94.4%; XGBoost: train 93.8%, test 93.4%; AdaBoost: train 87%, test 86.9%
[30]	RF, NB, CNN, LSTM (parallel and sequential training)	The dataset contains 54k training URLs and 12k testing URLs, sourced from the malicious and benign webpages dataset. Features include lexical, content-based, and network-level attributes	Evaluated using accuracy, precision, recall, *F*1-score, execution time, and speedup. NB achieved the highest accuracy (96.01%), while CNN and LSTM excelled in recall (100%)	NB: train 96.01%, test 96.01%; RF: train 95.14%, test 95.14%; CNN: train 95.13%, test 95.13%; LSTM: train 95.14%, test 95.14%

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Table 1: Continued.

Ref.	Model	Dataset	Performance evaluation	Training and testing accuracy
[31]	Logistic regression (LR), decision tree (DT), random forest (RF), support vector machine (SVM), Naïve Bayes (NB)	Fraud email dataset –11,925 labeled emails (spam/ham), Phishing email dataset – 82,486 emails from Enron, CEAS, Ling-Spam, SpamAssassin, Nazario corpora	Accuracy, Precision, Recall, *F*1-Score, Confusion Matrix analysis	Best model: random forest (RF): phishing dataset: accuracy 99.15%, fraud dataset: accuracy 98.72%. Other model performances (fraud dataset): logistic regression: 96.63%, Naïve Bayes: 94.53%, SVM: 97.66%, decision tree: 97.40%. Other model performances (phishing dataset): logistic regression: 96.65%, Naïve Bayes: 95.32%, SVM: 98.84%, decision tree: 98.14%
[32]	Random forest (RF)	Kaggle Phishing Website Dataset – 10,000 URLs (5000 phishing, 5000 legitimate) – Data collected via Selenium WebDriver between 2015 and 2017	Accuracy, precision, recall, *F*1-score, confusion matrix, ROC Curve/AUC	98.45% overall, legitimate: precision 97%, recall 98%, phishing: precision 99%, recall 97%
[33]	Rule-Based Classifier using: Domain Name Attributes, SEO Attributes, Security Scan Results (VirusTotal)	222 Phishing domain names from PhishTank	Classification accuracy rule impact analysis	99.1% overall accuracy (220/222 correct). Classes: phishing, suspicious, legitimate
[34]	j48 (C4.5 decision tree), Jrip (RIPPER), decision table, Naïve Bayes	Web Page Phishing Detection Dataset (11,430 URLs, 50% phishing, 87 features) Features from URL structure (56), page content (24), and external queries (7)	Accuracy, precision, recall, *F*1-score, MCC, ROC-AUC, PRC	JRip: 94.68%, j48: 94.59%; decision table: 92.10%; Naïve Bayes: 80.17%

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and recall, demonstrating its superior ability to handle sophisticated and ambiguous phishing attempts. This strategic approach ensures that our final framework is not only high-performing but also robust and generalizable across a wide range of real-world phishing threats.

4. Proposed Methodology

4.1. Overview of the Proposed Model. This phishing detection system is designed to identify whether online communications, such as emails, text messages, or URLs, are malicious (phishing) or legitimate, by leveraging both ML and DL approaches. It examines various characteristics of the messages, including their textual content and structural properties, to make accurate predictions.

The detection process follows a structured sequence of steps starting with gathering data, cleaning and preparing it, selecting the most relevant features, training the ML models, evaluating their performance, and finally, classifying new communications as safe or phishing attempts, as illustrated in Figure 4. To boost accuracy, the system combines traditional ML classifiers such as XGBoost, SVMs, and random forests with advanced DL models such as LSTM networks, CNNs integrated with LSTM (CNNs-LSTM), and BERT, which excels at understanding the language context. The entire approach operates in two main phases:

- Training phase: It is the phase where the system learns to recognize phishing patterns from labeled data.
- Real-time detection phase: It is the phase where it applies this knowledge to incoming communications instantly to detect and flag potential threats.
- This two-phase design enables the system to continuously improve while providing fast and reliable protection against phishing attacks.

4.1.1. Phase 1: Training and Testing

Step 1: Dataset preprocessing

Dataset compilation: The research begins with the compilation of a diverse dataset [25, 30] comprising three distinct sources: sms.csv, url.csv, and email.csv. This variety ensures a robust model capable of handling different communication mediums and phishing tactics. These datasets provide an excellent base for training and assessing ML and DL models.

4.1.1.1. Phishing Email Detection Dataset (Phishing_Email.csv). This dataset is available in [25] the Kaggle dataset, and we use it to analyze the email body to detect the phishing emails. This dataset reflects the growing concern of phishing emails and the need for improved detection systems. The email dataset consists of several key features for analysis. The primary component is the "Email Text," which comprises the body of the email and forms the basis for text-based analysis. Each email is also labeled with an "Email Type," categorized as either "Phishing" or "Safe," as shown in Table 2. A "Usability Score," ranging from 1 to 10, indicates the quality of the data for ML applications; in this instance, a score of 9.41 suggests good usability. The dataset is licensed under the GNU Lesser General Public License 3.0 and is updated several times a year.

4.1.1.2. SMS Phishing Dataset (Dataset_5971.csv). This dataset was released on Mendeley Data, created for detecting malicious SMS, including spam and smishing [30]. It was published by Sandhya Mishra and Devpriya Soni and consists of labeled SMS messages, which will be useful for any ML and pattern recognition task, as shown in Table 3. Each SMS message in the dataset is labeled with a classification of ham, spam, or smishing, indicating its category. The "Text" field contains the actual content of the SMS message. In addition, binary indicators are provided for the presence of URLs, email addresses, and phone numbers within the message, denoted as "URL/Email/Phone."

4.1.1.3. Kaggle Phishing and Legitimate URLs Dataset (new_data_urls.csv). The Kaggle Phishing and Legitimate URLs dataset [30] is balanced and extensive for URL-based phishing detection. It has URLs with binary labels, whether the domain is legitimate or phishing, thus providing a good basis for binary classification models. The dataset for URL classification includes "URL," which provides the domain or URL string to be analyzed. The "Status" field provides the classification label, using a binary system, where "0" indicates a phishing URL and "1" represents a legitimate URL, as shown in Table 4.

Preprocessing: Each dataset undergoes a crucial preprocessing stage. This involves the following:

Feature extraction: Identifying and isolating relevant features within the data that are indicative of phishing attempts or legitimate communication.

Feature vectorization: Transforming the extracted features into numerical vectors using established techniques such as TF-IDF, BERT, and Word2Vec. This numerical representation is essential for ML algorithms.

Dataset partitioning: The consolidated dataset is then divided into two subsets:

Training set: Used to train the learning algorithms and enable them to recognize patterns and characteristics associated with phishing.

Testing set: A held-out portion of the data used to evaluate the performance and generalization capabilities of the trained classifier.

Step 2: Model development and evaluation

Learning algorithm selection: Appropriate learning algorithms are chosen for the task, considering the nature of the data and the research objectives. This might involve exploring various classification algorithms such as logistic regression, SVMs, or DL models.

Classifier training: The selected classifier is trained using the prepared training dataset. This process involves optimizing the model's parameters to accurately distinguish between phishing and legitimate instances.

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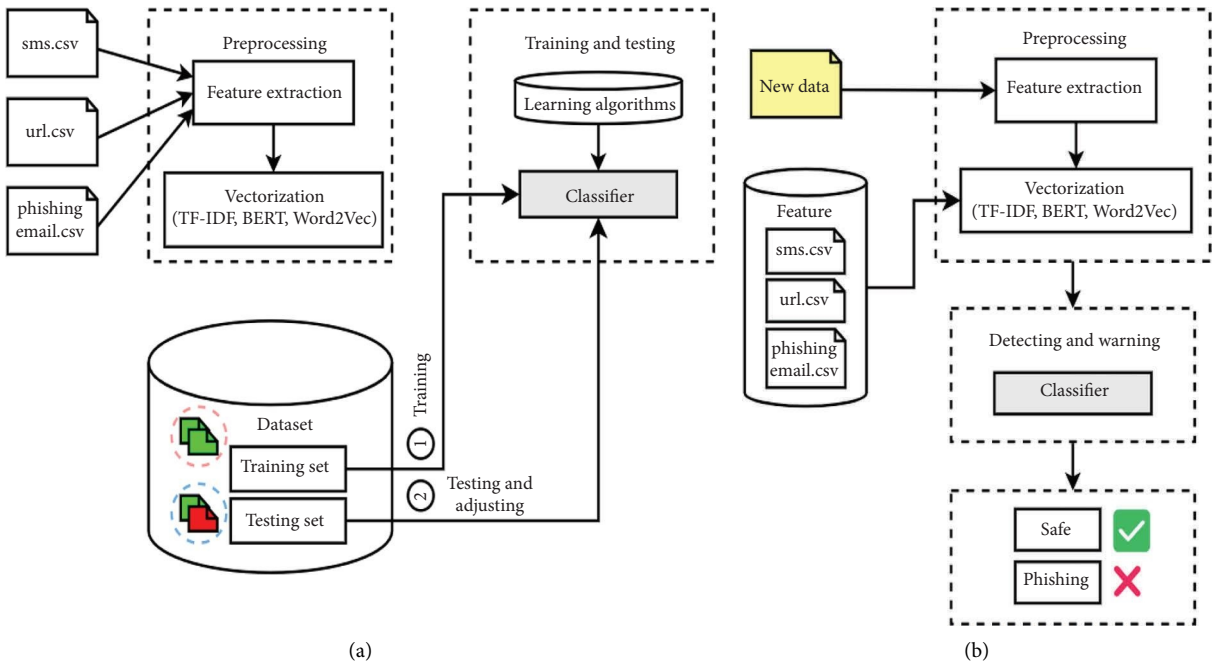


Figure 4: Proposed methodology.

Table 2: Several key features consist of the email phishing dataset.

Index	Email text	Email type
1324	Spanish terms I am looking for the Spanish translation...	Safe email
11038	conference in dc. July 20 - 21, 2005 4 th re...	Phishing email
11053	your online sales are low because you don _ t...	Phishing email
10804	---------- Yahoo! Groups Sponsor...	Safe email
18533	empty	Phishing email

Table 3: Several key features consist of the SMS phishing dataset.

Index	Label	Text
134	ham	New Theory: Argument wins d SITUATION, but loses the PERSON...
5286	ham	I promise to take good care of you, princess. I'll keep you happy always
445	ham	Ha... U just ate honey ar? So sweet...
556	ham	No calls...messages...missed calls...just missing your presence...
536	ham	I'll be at mu in like <#> seconds

Table 4: Several key features consist of the URLs phishing dataset.

Index	URL	Status
762544	carassai.com.ar	0
680300	tarnowiec.net.pl	0
290918	regiscerqueira.com.br/sd/gdoc/	0
660835	youtube.com/watch?v=UdZOL93r9R4	1
671138	k0lfeeuxjk.hikonebpp.com/lz03eiz6qp\n9pn1cwv66...	0

Testing and adjustment: The trained classifier is then evaluated using the held-out testing set. This provides insights into the model's performance on unseen data. Performance metrics such as accuracy, precision, recall, and *F1*-score are analyzed. Based on the results, adjustments are made to the model's parameters or preprocessing techniques to enhance its performance. This iterative process ensures an optimized and robust classifier.

Phase 2: Detection and warning

Step 3: Real-world application

New data input: In the second phase, the trained classifier is deployed to analyze new, unseen data. This represents the real-world application of the model for phishing detection.

Preprocessing (consistent with training): The new data undergo the same preprocessing steps (feature extraction and vectorization) as the training data, ensuring consistency and compatibility.

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Classification: The preprocessed data are fed into the trained classifier, which predicts whether the input is phishing or safe.

Step 4: Output and alert system

Detection and warning: Based on the classifier's output, the system generates an appropriate response. Legitimate inputs are marked as "Safe," while those identified as phishing are flagged with a "Phishing" warning. An effective alert mechanism is implemented to notify users of potential phishing attempts, enabling them to take appropriate precautions.

This two-phased methodology provides a comprehensive framework for developing and deploying a robust phishing detection system. The initial training and testing phase ensures the model's accuracy and reliability, while the subsequent detection and warning phase translates this capability into real-world protection for users. This systematic approach, grounded in established ML principles, offers a strong foundation for tackling the persistent challenge of phishing attacks.

4.2. Validity of the Proposed Model. To ensure the validity and reliability of our research, we implemented a rigorous experimental methodology that prioritized preventing data leakage and mitigating overfitting. Our approach was designed to produce robust and generalizable results, as evidenced by consistent high accuracy across all models.

A core principle of our methodology was the strict separation of data. We merged and then split our diverse datasets, spanning emails, URLs, and SMS, into distinct training and testing sets before any preprocessing began. This ensured that our models never had access to the test data during the training phase.

4.2.1. Vectorization and Scaling. For text vectorization (e.g., TF-IDF, Word2Vec, and BERT) and data scaling, we fit all methods exclusively on the training data. The resulting transformations were then applied to the test data. This practice prevented the models from "learning" from information they should not have seen.

4.2.2. Cross-Validation. We employed a stratified 5-fold cross-validation for all classical ML models (SVM, XGBoost, random forest, and KNN). This method ensures that each fold maintains a representative class balance, providing a more accurate assessment of a model's performance on unseen data.

4.2.3. DL Validation. For our DL models (LSTM, CNN-LSTM, GRU, BiGRU, and BERT), we used a dedicated hold-out validation set (typically 20% of the training data). We also incorporated early stopping with a set patience and applied dropout and regularization layers to prevent the models from becoming overly specialized to the training data.

4.2.4. Synthetic Data. When using techniques such as SMOTE or ADASYN to address class imbalance, we applied them only within the training folds during cross-validation. This precaution ensured that no synthetically generated examples could contaminate our validation or test sets.

While high accuracy can sometimes indicate overfitting, we are confident that our results are true. This precaution ensured that no synthetically generated examples could contaminate our validation or test results, reflecting the true performance of our models. This confidence stems from several factors as follows.

4.2.4.1. Diverse Datasets. Our models were trained and tested on three distinct and varied datasets, which provide a more realistic and rigorous test of their generalization capabilities than a single dataset would.

4.2.4.2. Multiclass Tasks. We performed multiclass classification, a more complex and challenging task rather than simple binary classification. This further demonstrates the model's ability to handle nuanced distinctions.

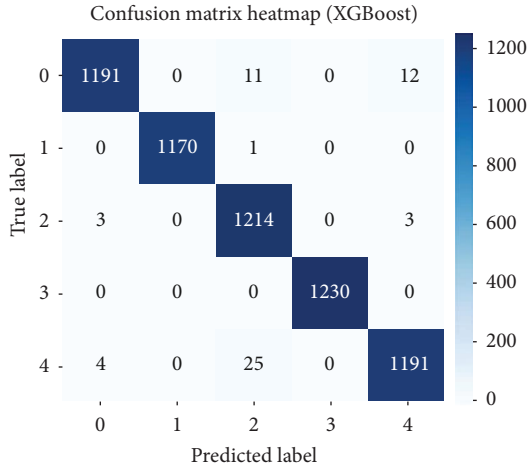
4.2.4.3. Consistency. The reported metrics (accuracy, precision, recall, and *F*1-score) were remarkably consistent across all cross-validation folds and models. This stability suggests a true and reliable performance rather than a fluke or a sign of overfitting.

5. Performance Analysis of the Proposed Model

5.1. XGBoost Model Evaluation. The XGBoost model demonstrated exceptional performance with an overall accuracy of 99.03%, as shown by the high concentration of correct predictions along the diagonal of its confusion matrix. While this is a very strong result, a closer look at the misclassifications reveals some of the model's limitations, particularly when dealing with the subtle linguistic nuances of phishing messages. Figure 5 shows that the XGBoost model performs strongly and consistently across all five target classes (0–4).

The model showed minor confusion between certain classes, highlighting the challenge of distinguishing between legitimate and malicious content that shares similar vocabulary. For instance, Class 2, which often includes ambiguous or "smishing" (SMS phishing) content, was sometimes misclassified as Class 0 (benign) or Class 4 (highly malicious). This suggests that short, service-related phrases in these messages can look a lot like legitimate communications, tricking the model's TF-IDF features. Similarly, benign messages (Class 0) were occasionally mistaken for phishing content (Classes 2 and 4). This likely happened because some legitimate alerts or transactional messages use urgent-sounding words that are also common in phishing attacks. The most frequent misclassification was between Class 4 (overtly malicious phishing) and Class 2, which suggests that the TF-IDF features struggle to differentiate between different types of phishing attacks when they use the same high-frequency words such as "verify" or "account." This shows that TF-IDF, while effective, cannot

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XGBoost accuracy: 0.9902559867877787

XGBoost classification report:

	Precision	Recall	*F*1-score	Support
0	0.99	0.98	0.99	1214
1	1.00	1.00	1.00	1171
2	0.97	1.00	0.98	1220
3	1.00	1.00	1.00	1230
4	0.99	0.98	0.98	1220
Accuracy			0.99	6055
Macro avg	0.99	0.99	0.99	6055
Weighted avg	0.99	0.99	0.99	6055

Figure 5: Confusion matrix and classification report of the XGBoost Model for phishing and safe classification.

fully capture the semantic context of words, it sees the words, but not the meaning behind them. In contrast, the model perfectly classified Class 3, suggesting this category contains messages with a very distinct structure or unique vocabulary that is not found in other classes, making it easier for the model to identify.

In summary, the high accuracy of the XGBoost model is a great achievement. However, the specific misclassifications reveal that its reliance on TF-IDF features makes it vulnerable to lexical overlap and semantic ambiguity. This highlights the need for future research to incorporate more advanced, context-aware language models to improve performance in these challenging, borderline cases.

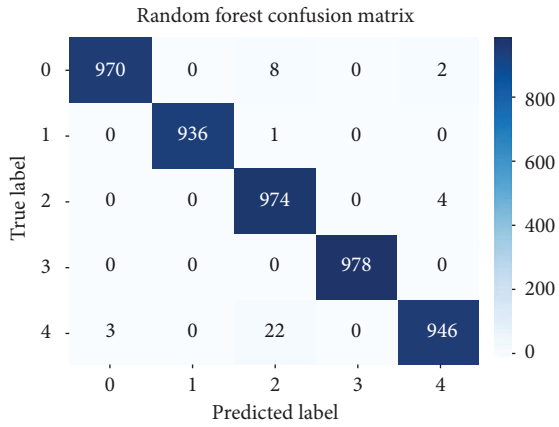
5.2. Random Forest Model Evaluation. Figure 6 illustrates the confusion matrix heatmap for the random forest classifier, showing strong overall predictive accuracy with a clear diagonal dominance across all five classes. However, a closer look at the off-diagonal cells reveals specific patterns of misclassification that point to limitations in representing lexical features. The random forest model struggled to differentiate between different types of phishing, with some Class 2 samples mistakenly labeled as Class 4 (email phishing). This confusion was due to the text-based features captured by TF-IDF being similar across these two classes. Class 4 samples were also misclassified, as they often used call-to-action phrases similar to Class 2. The model's difficulty in distinguishing Class 4 from other categories was due to its diverse nature and lack of a clear pattern. In addition, some legitimate emails in Class 0 were flagged as phishing due to word frequency, leading to false positives. However, the model achieved perfect precision and recall for Class 3, indicating highly consistent and unique patterns. The model's reliance on TF-IDF makes it vulnerable to semantic ambiguity and linguistic overlap, which calls for more advanced feature engineering.

5.3. SVM Evaluation. The SVM model effectively classifies phishing and legitimate messages into five classes, as demonstrated in Figure 7. Classes 1 and 3 achieve perfect precision and recall, indicating distinct textual or structural features. Class 0 (legitimate messages) also shows high accuracy, with 977 out of 980 samples correctly classified. However, three samples were misclassified as Class 4, likely due to formal alert-like phrases resembling phishing language. The main difficulty lies with Class 2, where 938 out of 978 instances are correctly identified, but 40 samples are misclassified into Classes 0 or 4. Class 2, often representing smishing or ambiguous phishing, shares significant linguistic overlap with benign and complex phishing messages, which TF-IDF's context-agnostic approach cannot resolve. SVM's linear kernel may also struggle to capture the subtle nonlinear boundaries needed to distinguish closely related classes with semantic overlap.

5.4. LSTM Model Evaluation. The LSTM model demonstrated strong classification performance in Figure 8, with a high degree of accuracy across all classes. It accurately classified Class 0 and Class 1, which have distinct language patterns, and Class 3, due to its unique sequential features. However, the model had some weaknesses, particularly in Class 2, where 15 messages were misclassified. These data suggest that the content in this category, such as smishing and ambiguous phishing attempts, shares linguistic overlap with legitimate and complex messages. Phrases such as "account update" or "verify identity" can appear in both legitimate and malicious contexts, making them difficult for the model to distinguish. The LSTM model's reliance on TF-IDF embeddings is a limitation, as it focuses on word frequency without understanding the true meaning or context of the words, making it difficult to accurately classify messages using similar vocabulary for different purposes.

5.5. CNN-LSTM Model Evaluation. Figure 9 displays the confusion matrix for the CNN-LSTM model, which has shown strong classification performance with an overall test accuracy of 99.50%. Class 1 (traditional phishing) is perfectly classified, indicating its ability to identify distinctive features such as urgency cues and impersonation

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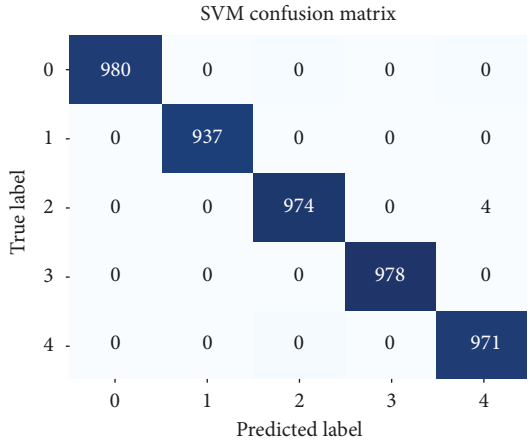


Random forest accuracy: 0.9917423616845582

Classification report:

	Precision	Recall	*F*1-score	Support
0	1.00	0.99	0.99	980
1	1.00	1.00	1.00	937
2	0.97	1.00	0.98	978
3	1.00	1.00	1.00	978
4	0.99	0.97	0.98	971
Accuracy			0.99	4844
Macro avg	0.99	0.99	0.99	4844
Weighted avg	0.99	0.99	0.99	4844

Figure 6: Confusion matrix and classification report of the random forest model for phishing and safe classification.

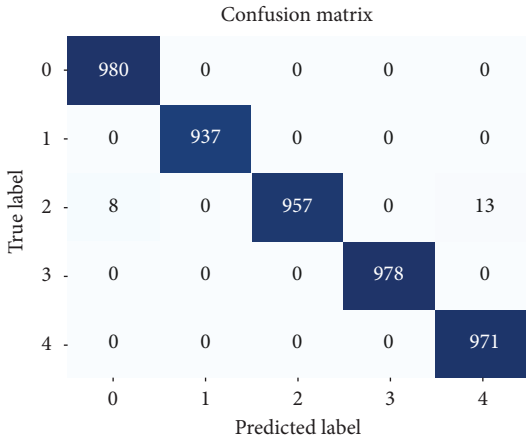


SVM accuracy: 0.9991742361684558

SVM classification report:

	Precision	Recall	*F*1-score	Support
0	1.00	1.00	1.00	980
1	1.00	1.00	1.00	937
2	1.00	1.00	1.00	978
3	1.00	1.00	1.00	978
4	1.00	1.00	1.00	971
Accuracy			1.00	4844
Macro avg	1.00	1.00	1.00	4844
Weighted avg	1.00	1.00	1.00	4844

Figure 7: Confusion matrix and classification report of the SVM model for phishing and safe classification.



LSTM test accuracy: 99.57%

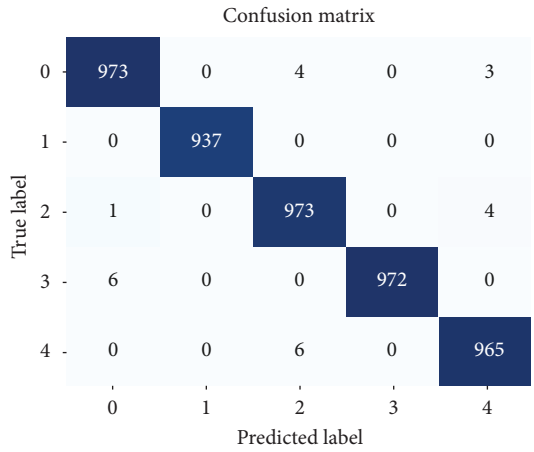
Classification report:

	Precision	Recall	*F*1-score	Support
0	0.99	1.00	1.00	980
1	1.00	1.00	1.00	937
2	1.00	0.98	0.99	978
3	1.00	1.00	1.00	978
4	0.99	1.00	0.99	971
Accuracy			1.00	4844
Macro avg	1.00	1.00	1.00	4844
Weighted avg	1.00	1.00	1.00	4844

Figure 8: Confusion matrix and classification report of the LSTM model for phishing and safe classification.

tactics. Class 0 (legitimate messages) shows minor misclassification into Classes 2 and 4, suggesting some benign messages contain call-to-action phrases or ambiguous tones resembling phishing characteristics. Class 2 (smishing or hybrid messages) is occasionally misclassified as Class 0 or 4, highlighting semantic overlap and lexical ambiguity. Class 3's few misclassifications into Class 0 indicate that certain phishing emails may resemble customer service communications. Class 4's errors are mostly misclassified as Class 2, indicating the difficulty in differentiating advanced phishing variants that share common tokens and delivery styles.

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CNN-LSTM accuracy: 99.50%

Classification report:

	Precision	Recall	*F*1-score	Support
0	0.99	0.99	0.99	980
1	1.00	1.00	1.00	937
2	0.99	0.99	0.99	978
3	1.00	0.99	1.00	978
4	0.99	0.99	0.99	971
Accuracy			1.00	4844
Macro avg	1.00	1.00	1.00	4844
Weighted avg	1.00	1.00	1.00	4844

Figure 9: Confusion matrix and classification report of the CNN-LSTM model for phishing and safe classification.

5.6. KNN Model Evaluation. The KNN classifier achieves high accuracy across all classes after balancing the dataset using ADASYN. However, Class 2 shows a noticeable recall drop to 0.96 due to 32 misclassifications out of 730 samples as shown in Figure 10. These errors mainly involve samples being mislabeled as Class 0 and Class 4, with a few misclassifications into Classes 1 and 3. These mistakes likely arise from lexical and structural overlaps between classes, such as Class 0 and Class 2 sharing benign or ambiguous language. The minor confusion involving Classes 1 and 3 suggests that current TF-IDF feature representations have limited capacity to fully distinguish the subtle boundaries of Class 2. This suggests a need for richer semantic modeling approaches to improve classification accuracy in challenging cases.

5.7. GRU Model Evaluation. The confusion matrix shows that in Figure 11, the GRU model achieves a high overall accuracy of 99.50%, though some minor misclassifications remain, mainly between closely related classes. Class 2 (phishing SMS) experiences 18 misclassifications, with 16 samples labeled as Class 4 (phishing email) and Class 2 as Class 0 (safe SMS). This reflects overlapping linguistic cues, such as security prompts, call-to-action phrases, and service-related keywords, that TF-IDF features alone cannot distinguish. Despite GRU's ability to model sequences, it struggles to resolve these semantic overlaps fully without richer contextual embeddings.

In contrast, Class 3 (legitimate URL) is perfectly classified, likely due to its distinct structural characteristics that make it separable. Overall, the results highlight the GRU model's strong potential while emphasizing the importance of enhanced feature representations, such as BERT or other contextual embeddings, to better differentiate phishing variants that share subtle textual similarities.

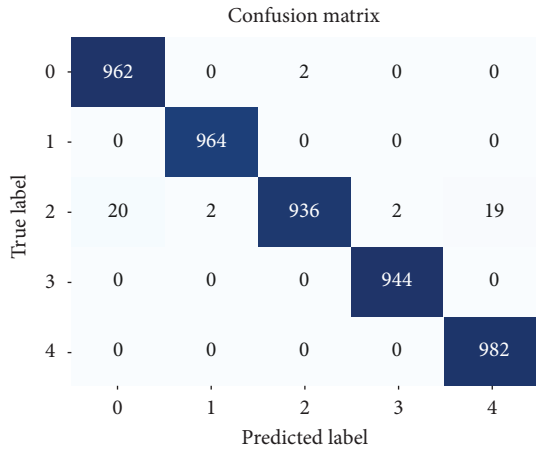
5.8. BiGRU Model Evaluation. The BiGRU model's confusion matrix heatmap in Figure 12 shows strong classification reliability, with Class 1 and Class 3 achieving perfect classification. Class 0 (safe SMS) shows minimal misclassifications into Classes 2 and 4, likely due to shared linguistic features. Class 2, representing subtle or ambiguous phishing attempts, has a few misclassifications into both legitimate and spoofed phishing classes. Class 4 exhibits minor confusion with legitimate content, indicating vulnerability to sophisticated phishing tactics disguised as malicious intent. The model's robustness is highlighted, with Class 2 requiring further integration of richer contextual embeddings or transformer-based architectures for improved disambiguation and reduced errors.

5.9. BERT Model Evaluation. The confusion matrix for the BERT model highlights its strong performance in distinguishing phishing from safe messages, with an overall accuracy of 98.09% and both macro and weighted *F*1-scores at 0.98. The model successfully classified 1645 out of 1688 phishing samples (97% recall) and 3182 out of 3233 safe samples (98% recall), demonstrating balanced and reliable results across both categories, as shown in Figure 13.

While false positives, safe emails mistakenly flagged as phishing, pose minor user inconvenience, the 43 false negatives, where phishing emails were misclassified as safe, represent a more significant security risk. This underlines the critical need to minimize such errors in cybersecurity applications. The heatmap visualization not only emphasizes the model's strong predictive confidence through a concentrated diagonal pattern but also reflects the difficulty of classifying borderline cases where linguistic and structural features overlap subtly between classes.

This comparison shown in Table 5 indicates that most of our proposed models are as good as or even better than the state-of-the-art models in the literature, indicating their high potential for phishing detection. Among them, the SVM model achieved the best accuracy of 99.91%, the best model for phishing detection. Furthermore, DL models, XGBoost (99.03%), random forest (98.99%), LSTM (99.57%), hybrid (CNN-LSTM) (99.50%), KNN (99.07%), GRU (99.56%), and BiGRU (99.75%), were also state-of-the-art, which demonstrates that they are capable of learning complicated patterns and sequential dependencies in phishing data. In addition, the BERT model (98.09%), known for its advanced

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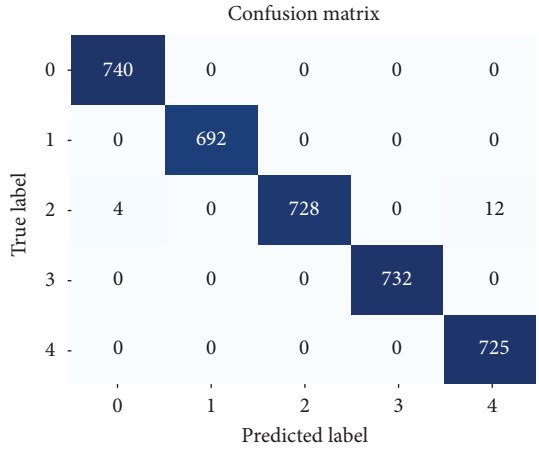


KNN accuracy: 0.9906890130353817

Classification report:

	Precision	Recall	*F*1-score	Support
0	0.98	1.00	0.99	964
1	1.00	1.00	1.00	964
2	1.00	0.96	0.98	979
3	1.00	1.00	1.00	944
4	0.98	1.00	0.99	982
Accuracy			0.99	4833
Macro avg	0.99	0.99	0.99	4833
Weighted avg	0.99	0.99	0.99	4833

Figure 10: Confusion matrix and classification report of the KNN model for phishing and safe classification.

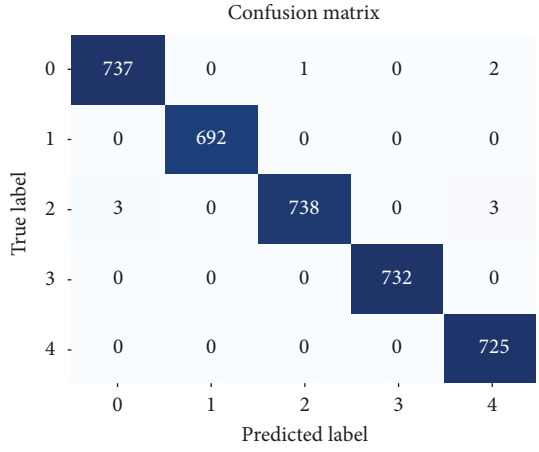


GRU accuracy: 99.56%

Classification report:

	Precision	Recall	*F*1-score	Support
0	0.99	1.00	1.00	740
1	1.00	1.00	1.00	692
2	1.00	0.98	0.99	744
3	1.00	1.00	1.00	732
4	0.98	1.00	0.99	725
Accuracy			1.00	3633
Macro avg	1.00	1.00	1.00	3633
Weighted avg	1.00	1.00	1.00	3633

Figure 11: Confusion matrix and classification report of the GRU model for phishing and safe classification.



BiGRU accuracy: 99.75%

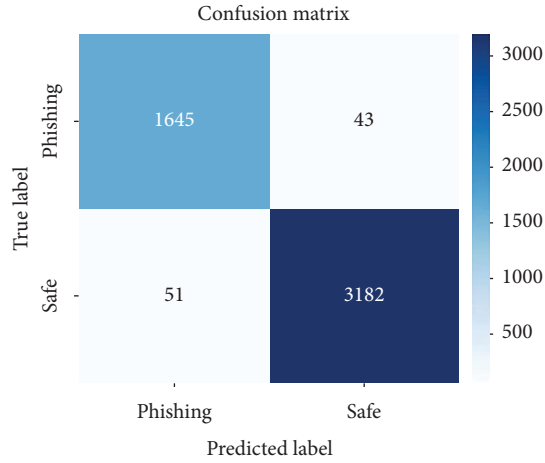
Classification report:

	Precision	Recall	*F*1-score	Support
0	1.00	1.00	1.00	740
1	1.00	1.00	1.00	692
2	1.00	0.99	1.00	744
3	1.00	1.00	1.00	732
4	0.99	1.00	1.00	725
Accuracy			1.00	3633
Macro avg	1.00	1.00	1.00	3633
Weighted avg	1.00	1.00	1.00	3633

Figure 12: Confusion matrix and classification report of the BiGRU model for phishing and safe classification.

natural language understanding, also exhibited competitive accuracy, confirming its excellence in phishing detection tasks. These results highlight the significance of DL models, feature engineering, and fine-tuned preprocessing methods in maximizing phishing detection performance. The findings indicate that the combination of SVM, BiGRU, LSTM, GRU,

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BERT accuracy: 0.9808981914245072

Classification report:

	Precision	Recall	*F*1-score	Support
0	0.97	0.97	0.97	1688
1	0.99	0.98	0.99	3233
Accuracy			0.98	4921
Macro avg	0.98	0.98	0.98	4921
Weighted avg	0.98	0.98	0.98	4921

Figure 13: Confusion matrix and classification report of the BERT model for phishing and safe classification.

Table 5: Performance comparison of phishing detection techniques.

Ref	Email	SMS	URL	Accuracy
[14]	✓	✓	✓	K-means 56.98%
[15]	×	×	✓	CNN: 98.77%, RNN: 97.4%, random forest: 94.26%
[16]	×	×	✓	CNN: 99.2%, LSTM-CNN: 97.6%, LSTM: 96.8%
[17]	×	×	✓	NN: 90.23%, Naïve Bayes: 92.97%, AdaBoost: 95.43%
[18]	×	×	✓	Random forest: 98.3%, SVM: 98.3%
[25]	✓	×	✓	CNN: 98.7%, LSTM: 97.5%, hybrid models: 99.1%
[26]	✓	×	×	SVM: 99.62%, LSTM: 98%, NB: 97%
[29]	×	×	✓	Naïve Bayes: 96.01%, random forest: 95.14%, CNN-LSTM: 95.13%
Our training model	✓	✓	✓	XGBoost: 99.03%, random forest: 98.99%, SVM: 99.92%, LSTM: 99.57%, hybrid (CNN-LSTM): 99.50%, KNN: 99.07%, GRU: 99.56%, BiGRU: 99.75%, BERT: 98.09%.

and hybrid models with BERT can make a solid foundation for building robust and effective phishing detection systems for real-world cybersecurity environments.

6. Conclusion and Future Work

This study presents a comprehensive evaluation of advanced ML and DL models for phishing detection, emphasizing the development and assessment of robust methodologies to combat the evolving landscape of cyber threats. By leveraging diverse datasets comprising emails, URLs, and SMS messages, the proposed framework achieved exceptional accuracy, with SVM (99.91%) leading the way, followed by BiGRU (99.75%), LSTM (99.57%), and hybrid (CNN-LSTM) (99.50%) models, which set new benchmarks. These findings highlight the effectiveness of hybrid architectures in capturing both spatial and sequential patterns, making them highly suitable for detecting sophisticated phishing attempts. The implementation of advanced preprocessing techniques, such as TF-IDF vectorization, feature scaling, and class balancing with SMOTE and ADASYN, significantly improved model training and generalization, ensuring high detection accuracy across different phishing formats. In addition, traditional models such as SVM demonstrated competitive performance with minimal computational overhead, reinforcing their suitability for resource-constrained applications such as browser extensions and mobile security solutions. These results validate the effectiveness of DL-driven phishing detection systems, paving the way for future enhancements through domain-specific pretraining, adaptive learning techniques, and real-time threat detection frameworks.

Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

Ethics Statement

The authors have nothing to report.

Consent

The authors have nothing to report.

Disclosure

All authors have read and agreed to the published version of the manuscript.

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Conflicts of Interest

The authors declare no conflicts of interest.

Author Contributions

Conceptualization: Md. Humayun Kabir, Mohammad Zibran Hasan, and Mehraj Ibne Halim. Methodology: Md. Humayun Kabir, Mehraj Ibne Halim, Mohammad Zibran Hasan, and Mohammad Nadib Hasan. Formal analysis: Mehraj Ibne Halim, Mohammad Zibran Hasan, Ahmad, Mohammad Nadib Hasan, and Hasibul Hasan. Investigation: Mehraj Ibne Halim and Mohammad Zibran Hasan. Resources: Md. Humayun Kabir, Mehraj Ibne Halim, Mohammad Zibran Hasan, and Hasibul Hasan. Writing–original draft preparation: Md. Humayun Kabir, Mehraj Ibne Halim, Mohammad Zibran Hasan, and Hasibul Hasan. Writing–review and editing: Md. Humayun Kabir, Mohammad Nadib Hasan, Hassan Jaki, Ahmad, and Hasibul Hasan. Supervision: Md. Humayun Kabir and Mohammad Nadib Hasan. Project administration: Md. Humayun Kabir, Hassan Jaki, Ahmad, and Mohammad Nadib Hasan. Funding acquisition: Md. Humayun Kabir and Hasibul Hasan.

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